Cross-fitted marginal-estimand PC/TMLE analysis

Complete methodology, inference, multiplicity adjustment, and saved study figures

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Purpose. This report documents the marginal-estimand branch for the fixed 12-gene panel and the all-999-gene study. The target is a symmetric log interaction of the marginal joint distribution of a selected gene set. It is an association among binary expression states, not a causal effect, direct regulatory claim, or biological mechanism. The report is self-contained and uses plain language alongside the mathematics; it does not append raw Markdown source notes.

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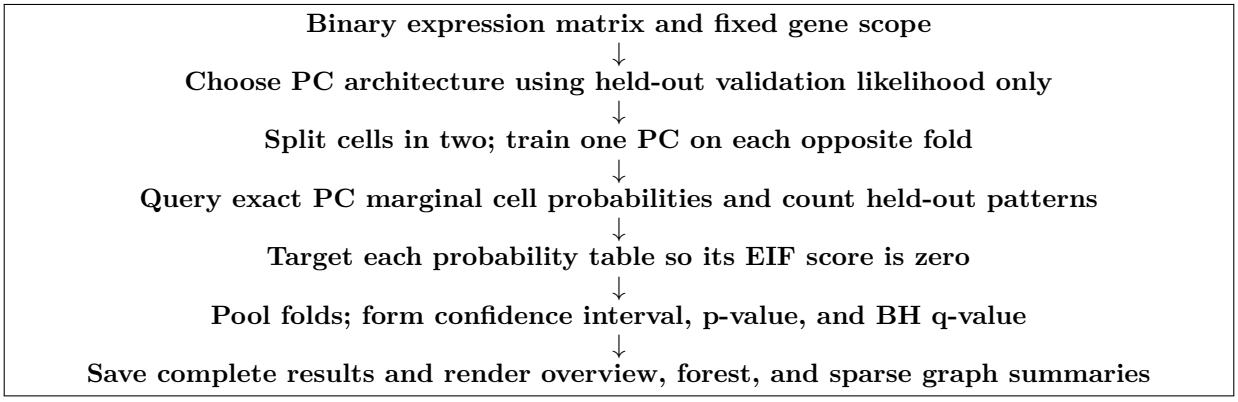
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1 Analysis pipeline at a glance

The same sequence drives the 12-gene and 999-gene analyses. The 999-gene study uses a special pair cache and screens before higher orders, but it estimates the same marginal target and uses the same targeting and inference steps.



Every requested 12-gene-panel set is evaluated. At 999 genes, every pair is feasible because of the exact pair cache; orders three and above are selected from pairwise results and are explicitly exploratory.

1.1 Reproducible commands and saved artifacts

Run from the repository root:

```
python sweep_pc_architectures.py -analysis 12
python train_cross_fitted_pcs.py -analysis 12
python cross_fitted_interactions.py -analysis 12 -interaction-order r
```

Replace 12 by 999 as appropriate. The 999-gene pairwise command can include `-rebuild-cache`. Result files, checkpoints, and the pair cache carry a SHA-256 fingerprint of ordered gene names and binary values. This fails safely if data, column order, or scope changes. Result tables keep the selected gene set, PC plug-in and targeted effects, standard error, confidence interval, p-value, BH q-value, support flags, and smallest observed count/model probability.

2 Data, PC fitting, and cross-fitting

The source CSV contains 11,950 cells and 1,000 numeric columns; its constant unnamed first metadata column is removed, leaving 999 binary named gene columns. The 12-gene panel has a fixed analysis order:

`Spc24, Gmnn, Prim2, Gins2, Cks2, Cdca8, Pbk, Ube2c, Ccna2, Apitd1, Hells, Mcm6.`

No cells or genes are filtered. A SHA-256 fingerprint of ordered gene names and binary values is stored with every checkpoint and result. Reduced-sample runs deliberately change this fingerprint so they cannot reuse full-data artifacts.

Architecture selection is separate from inference. With seed 0, 9,560 cells are used for selection training and 2,390 for validation; candidates are ranked by mean validation negative log likelihood. The 12-gene selection evaluates 39 combinations of four PC architectures, component counts, and learning rates, selecting a 32-component mixture-Chow–Liu model (validation NLL 4.081627). A mixture tree PC has the form

$$p(x) = \sum_z \pi_z p(x_{\text{root}} \mid z) \prod_{j \neq \text{root}} p\{x_j \mid x_{\text{parent}(j)}, z\}.$$

For tree candidates, the Chow–Liu structure is learned from selection-training cells only using smoothed mutual information and a maximum-spanning tree. No interaction effect, p-value, or q-value is used in selection.

The selected specification is then fitted separately across a two-fold split of 5,975 held-out cells per fold. PC_1 learns from fold 2 and evaluates fold 1; PC_2 learns from fold 1 and evaluates fold 2. Thus every result uses a PC that did not train on the evaluated cell. Fold estimates are combined using held-out-fold sample-size weights and EIF moments are pooled. For 12 genes, final PCs use the 32-component mixture-Chow–Liu architecture, 20 epochs, 128-cell minibatches, learning rate 0.02, weight decay 10^{-5}, and CPU. The 999-gene final configuration is retained in its guide and uses a 32-component latent-class residual tree.

3 Marginal estimand and exact probability/count calculation

For selected gene set S of order r, let $a \in \{0,1\}^r$ denote a binary expression pattern and define $s(a) = (-1)^{r-\sum_j a_j}$. The marginal interaction estimand is

$$\theta_r(p) = \sum_{a\in\{0,1\}^r} s(a) \log p(X_S = a).$$

For a pair this reduces to the log odds ratio

$$\theta_2(p) = \log \frac{p_{00}p_{11}}{p_{01}p_{10}},$$

where $p_{ab} = p(X_i = a, X_j = b)$. Zero means no interaction on this scale; a positive or negative value is positive or negative marginal association. For larger sets, exponentiating produces an alternating cross-product ratio.

The PC obtains every selected-set pattern probability exactly by tree message passing: it fixes selected states, sums over all other tree nodes, and averages over mixture components. It does not simulate cells. Held-out pattern counts are created in bulk by binary encoding plus `torch.bincount`. In the all-999-gene pairwise study, a special exact cache holds $p_i = p(X_i = 1)$, $p_{11,ij} = p(X_i = 1, X_j = 1)$, observed $n_i, n_{11,ij}$, and fold sizes for each of 498,501 unordered pairs. The other cells follow from

$$\begin{aligned} p_{00} &= 1 - p_i - p_j + p_{11}, & n_{00} &= n - n_i - n_j + n_{11}, \\ p_{01} &= p_j - p_{11}, & n_{01} &= n_j - n_{11}, \\ p_{10} &= p_i - p_{11}, & n_{10} &= n_i - n_{11}. \end{aligned}$$

This analytic cache makes exhaustive pairs practical without enumerating a 2^{999} state space; it does not generalize to exhaustive higher orders.

4 TMLE update, score, and EIF

Within each held-out fold and target, the initial PC probability table $p_0(a)$ is updated in the signed fluctuation submodel

$$p_\epsilon(a) = p_0(a) + \epsilon s(a).$$

The feasible ϵ interval maintains strict positivity of all cells. For a pair with $s = (+1, -1, -1, +1)$,

$$\max(-p_{00}, -p_{11}) < \epsilon < \min(p_{01}, p_{10}).$$

After moving endpoints slightly inward, 64 vectorized bisection iterations solve the empirical EIF score equation

$$\sum_a p_n(a) \frac{s(a)}{p_\epsilon(a)} = 0.$$

TMLE therefore chooses a probability-table update with empirical EIF mean zero; it does *not* minimize the EIF. At the targeted table p^*, the centered EIF is

$$H(a) = \frac{s(a)}{p^*(a)} - \mathbb{E}_{p^*}\left\{ \frac{s(A)}{p^*(A)} \right\}.$$

The updated fold log interactions are combined with held-out sample-size weights. This targeting, exact PC probability calculation, and bulk count construction are vectorized across targets and folds.

5 Unregularized support, uncertainty, p-values, and q-values

Two result states are intentionally distinct. A target is *targeted* when its initial table is finite and strictly positive and the bisection equation has a finite root in both folds. It is *supported for unregularized inference* only when it is targeted and, in each fold, all 2^r observed patterns meet `minimum-cell-count`, all initial PC probabilities meet `minimum-probability`, and the table is finite. Targeted-but-unsupported estimates remain descriptive but get no SE, CI, p-value, or q-value. The support rules are gates: no pseudocount, continuity correction, probability floor, ridge penalty, or variance regularization alters counts/probabilities.

Let N be the pooled held-out sample size. The pooled EIF standard error is

$$\widehat{\mathrm{SE}}(\theta) = \sqrt{\frac{\sum_i H_i^2 - (\sum_i H_i)^2/N}{(N-1)N}}.$$

The table reports $\theta \pm 1.959964\,\widehat{\mathrm{SE}}$, and the two-sided normal p-value for $H_0 : \theta = 0$ is

$$p_h = 2\Phi\Big(-\Big|\theta_h/\widehat{\mathrm{SE}}_h\Big|\Big).$$

For the supported family of m hypotheses, sort $p_{(1)} \le \cdots \le p_{(m)}$. This branch reports Benjamini–Hochberg (BH) adjusted p-values:

$$q_{(i)}^{\mathrm{BH}} = \min_{j \ge i} \min\left\{1, \frac{m}{j} p_{(j)}\right\}; \qquad p_{(k)} \le \frac{k\alpha}{m}.$$

The first expression is returned to each original target; the second is the BH selection condition at FDR α. Thus $q_h \le \alpha$ says the target is selected when BH is run at α, not that q_h is a posterior probability or a probability that an edge is false. BH's usual FDR guarantee requires valid null p-values plus independence or positive regression dependence.

For comparison, Benjamini–Yekutieli (BY), used on the conditional-estimand branch and suitable for arbitrary dependence, inserts the harmonic factor $c(m) = \sum_{\ell=1}^{m} 1/\ell$:

$$q_{(i)}^{\mathrm{BY}} = \min_{j \ge i} \min\left\{1, \frac{m\,c(m)}{j} p_{(j)}\right\}; \qquad p_{(k)} \le \frac{k\alpha}{m c(m)}.$$

6 How to read every plotted result

The tall labelled comparison figure is the complete panel record: it gives every requested target a row, rather than pre-screening it away. A circle is the PC plug-in estimate; an orange x is the targeted estimate; an orange interval is reported only for targets supported for unregularized inference; and a green star, when available, is a full-data empirical comparison. The dashed vertical zero line is no marginal interaction. The smaller overview, forest, network, and hypergraph figures are readable summaries of this full result table.

Overview (effect and evidence). Each finite supported target is one point. Its horizontal position is its TMLE log interaction, with red for non-negative and blue for negative effects. Its vertical position is now $-\log_{10}(p_h)$, the *unadjusted* two-sided p-value, capped at 50 only for drawing. Thus heights 1.301, 2, and 3 correspond to p-values 0.05, 0.01, and 0.001. The horizontal line at 1.301 is a nominal unadjusted p-value 0.05 reference, not a BH false-discovery-rate boundary. Dashed vertical lines at ± 0.25 mark a practical effect-size reference. Here, "evidence" means lower compatibility with an exact-zero interaction under the normal/EIF approximation; it is not the probability that a specific association is real, false, causal, or biologically useful. The BH q-value is still saved in the table and is used for selecting the smaller summaries.

Forest plot (selected effect estimates). Form $D = \{h : \text{supported finite inference},\ q_h^{\mathrm{BH}} \le 0.05,\ |\theta_h| \ge 0.25\}$. The q-value cut controls the intended multiple-testing selection procedure; the effect-size cut avoids using tiny but precisely estimated effects as visual highlights. If D is empty, the plot falls back to all supported finite targets, so a fallback is a ranking, not a discovery set. It takes no more than 25 negative and 25 non-negative effects independently, ordered from the most negative and most positive. A dot is the TMLE estimate; the bar is its 95% Wald interval; black zero is no interaction. A confidence interval that crosses zero is not removed by an extra rule; the stated BH and magnitude cuts are the only default display cuts.

Network and hypergraph (sparse structure). Pair graphs begin with the same D (or fallback set), sort by $|\theta_h|$, then greedily retain at most 25 edges while every gene has degree below four. A significant-looking pair can therefore be absent simply because one endpoint already has four stronger displayed neighbours. Fewer than 25 edges means that candidates ran out or degree caps prevented more. Red/blue shows sign, width is relative magnitude, and node size is retained degree. Disconnected components are laid apart for legibility; they do not prove biological independence. At higher orders, at most eight strongest screened gene sets are outline polygons. Shared or overlapping polygons merely share genes and have no extra statistical meaning.

7 Results: 12-gene study

The saved full-panel run used `minimum-cell-count=1`, `minimum-probability=1e-12`, target batches of 512, and 64 bisections. It trained once (2.13 s complete command) and then ran each full order; total fresh sequential time was 23.26 s. The following complete-command timings include loading, queries, counts, targeting, inference, result saving, and figures.

Order	Targets	Interaction command	Fresh fit + one order
2	66	1.60 s	3.73 s
3	220	2.01 s	4.14 s
4	495	2.95 s	5.08 s
5	792	4.47 s	6.60 s
6	924	5.28 s	7.41 s
7	792	4.82 s	6.95 s

The saved plots below provide the available overview, forest, and network/hypergraph results for every completed panel order. Their overview vertical axis is $-\log_{10}(p)$; their forest, network, and hypergraph selection still uses BH q-values plus the stated effect-size screen. The full tall labelled comparison figures remain in the study directory as the complete, unscreened panel record.

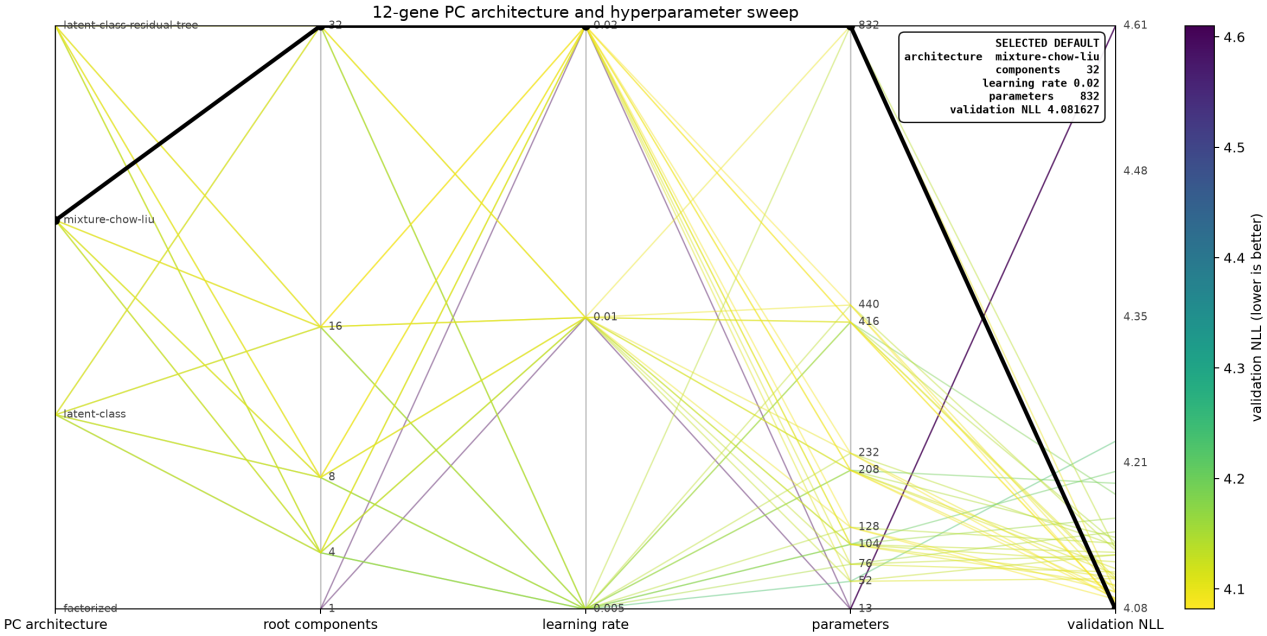


Figure 1: 12-gene PC architecture sweep retained for provenance.

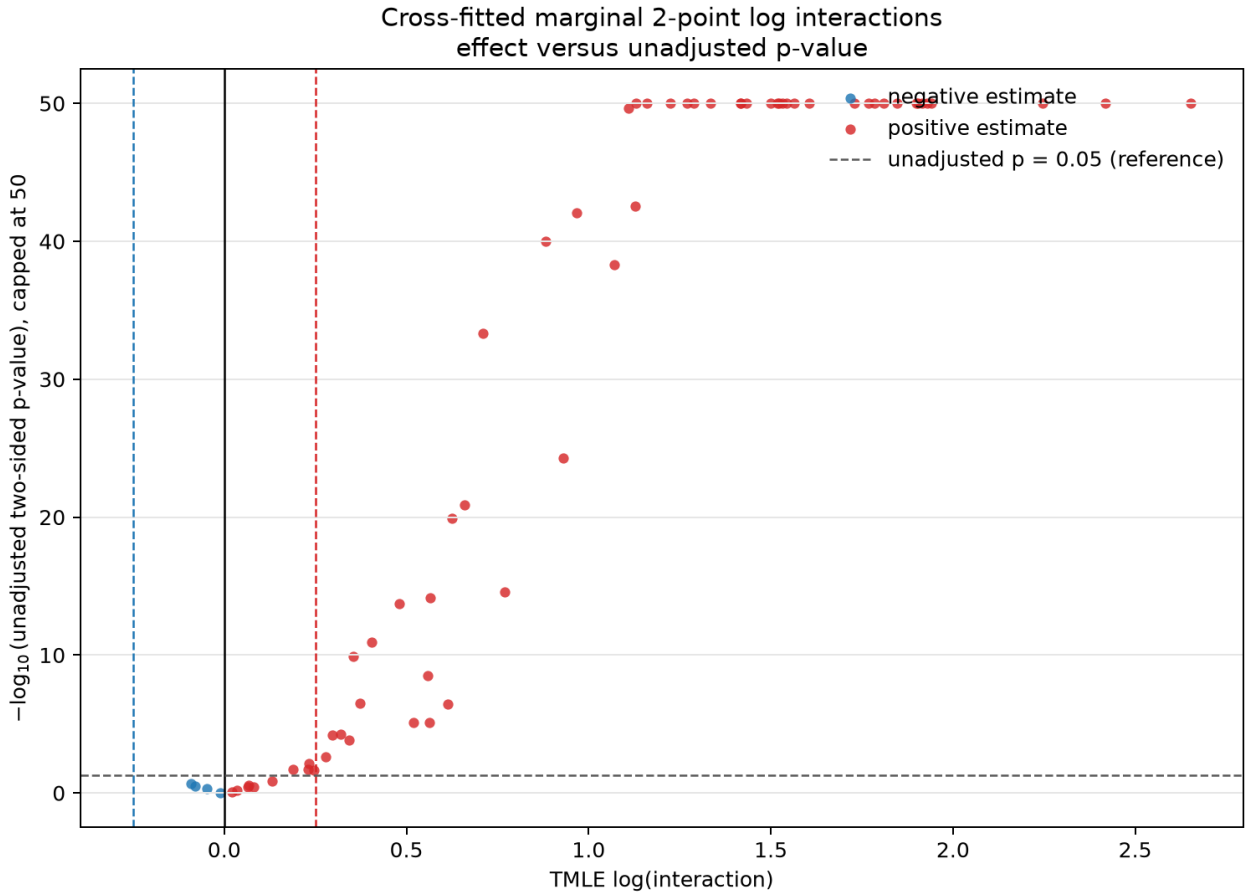


Figure 2: 12-gene marginal interactions, order 2: overview.

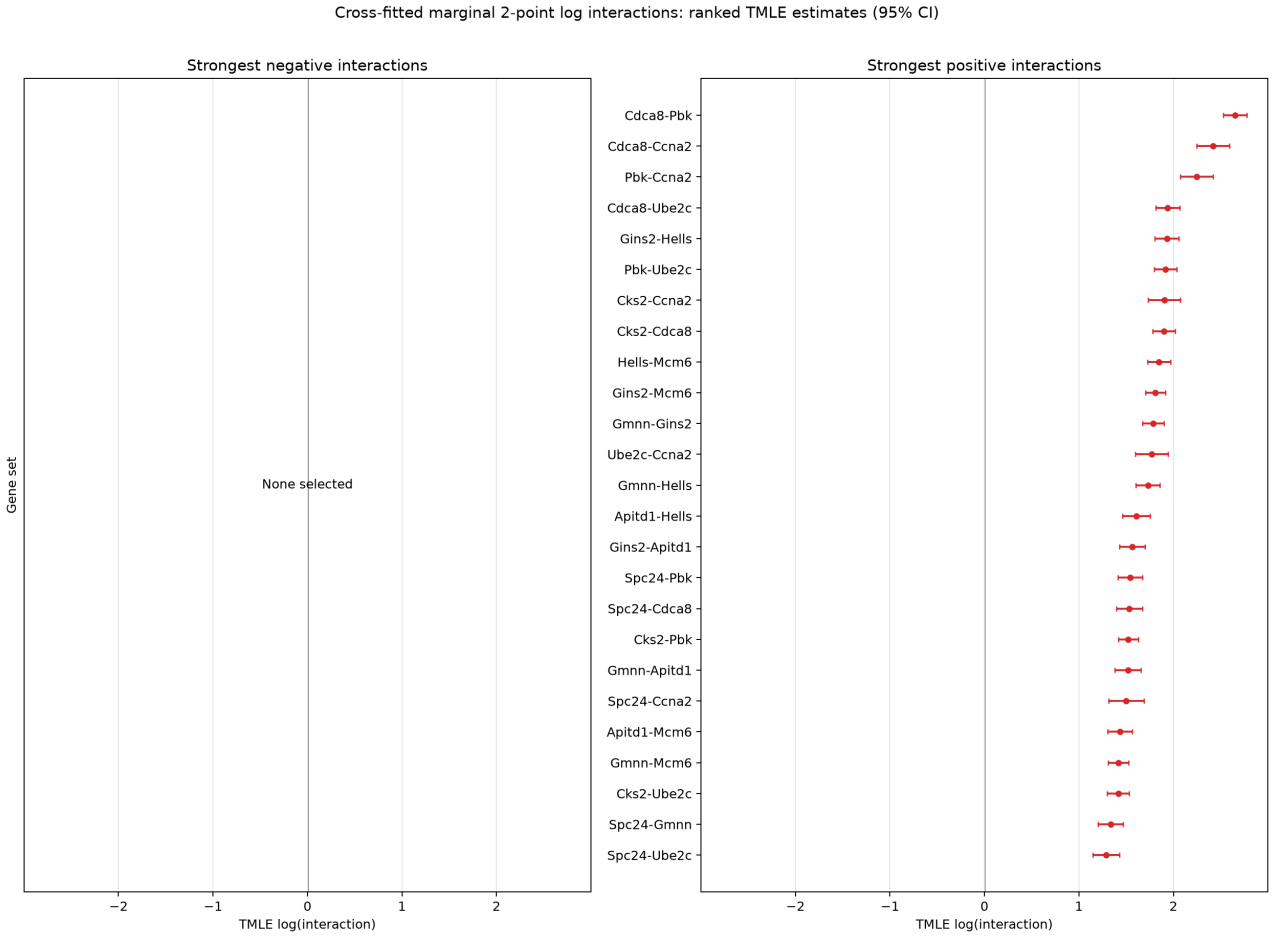


Figure 3: 12-gene marginal interactions, order 2: forest summary.

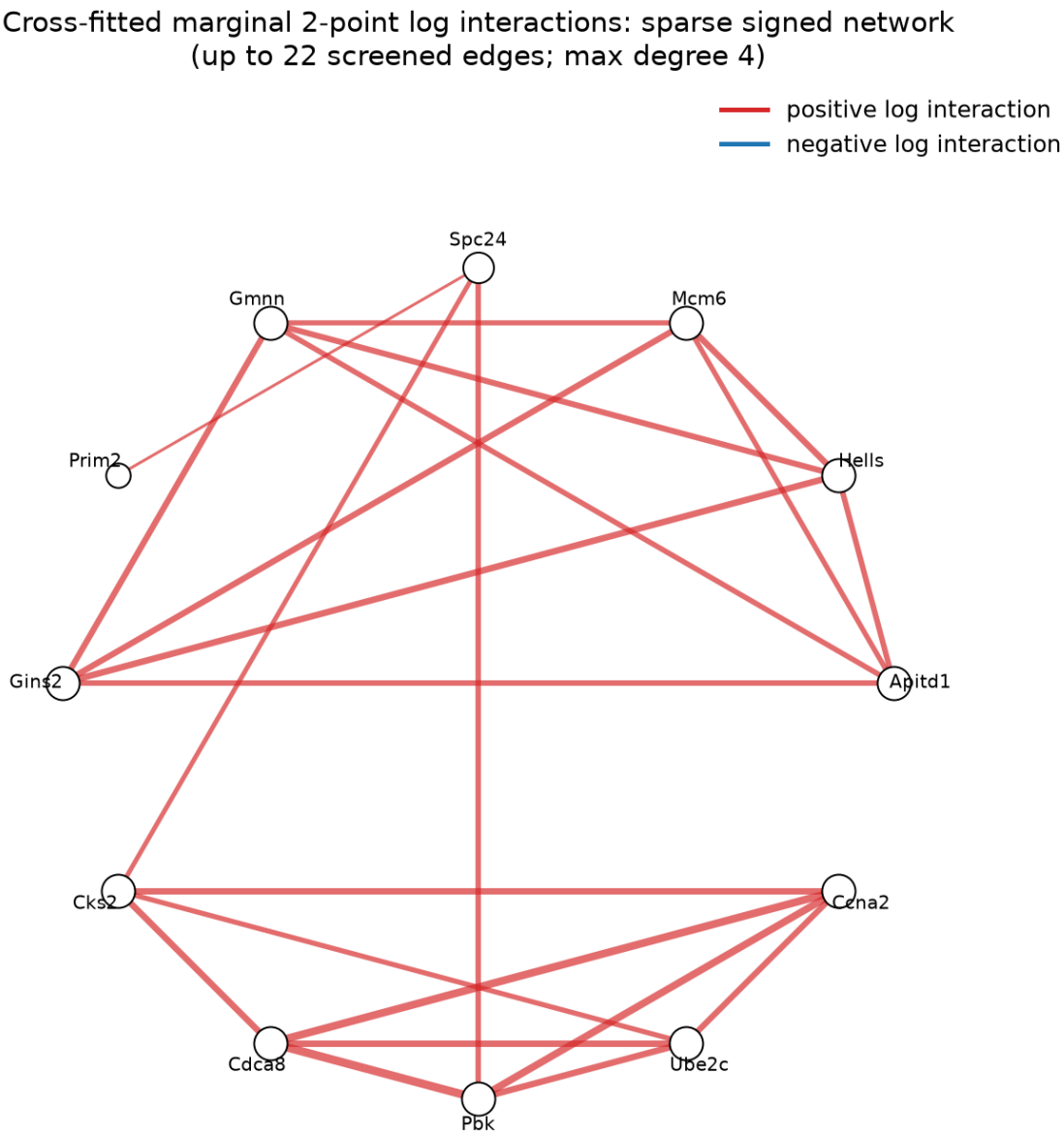


Figure 4: 12-gene marginal interactions, order 2: network.

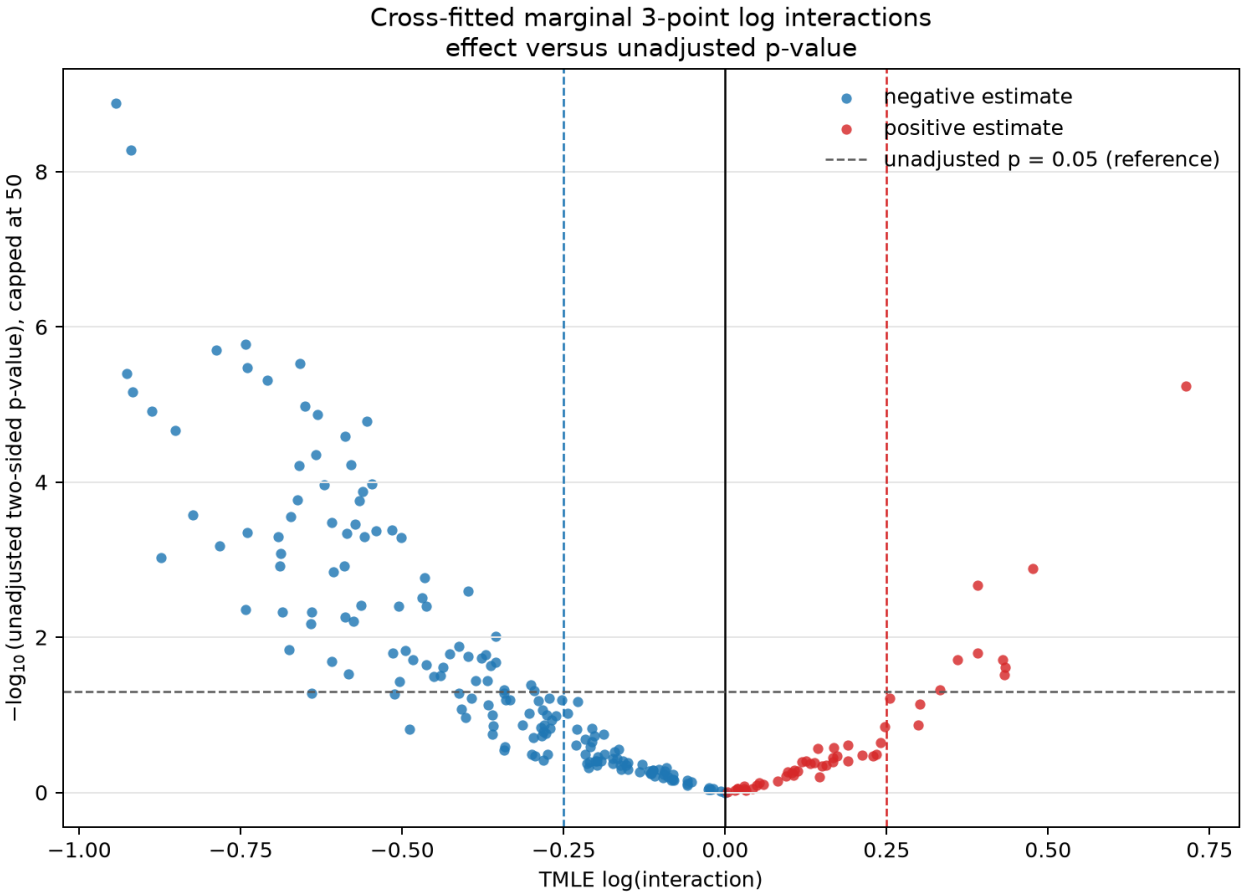


Figure 5: 12-gene marginal interactions, order 3: overview.

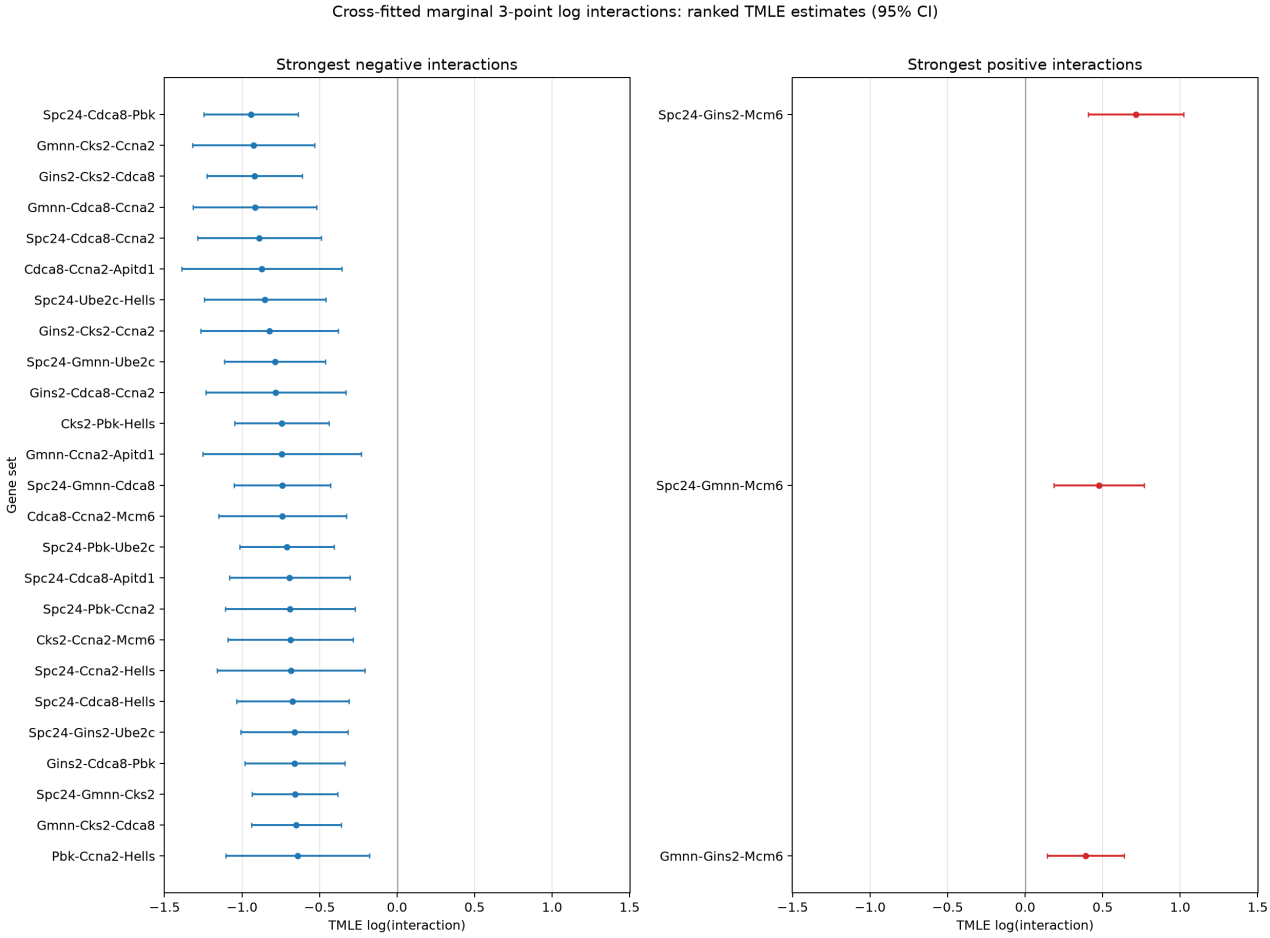


Figure 6: 12-gene marginal interactions, order 3: forest summary.

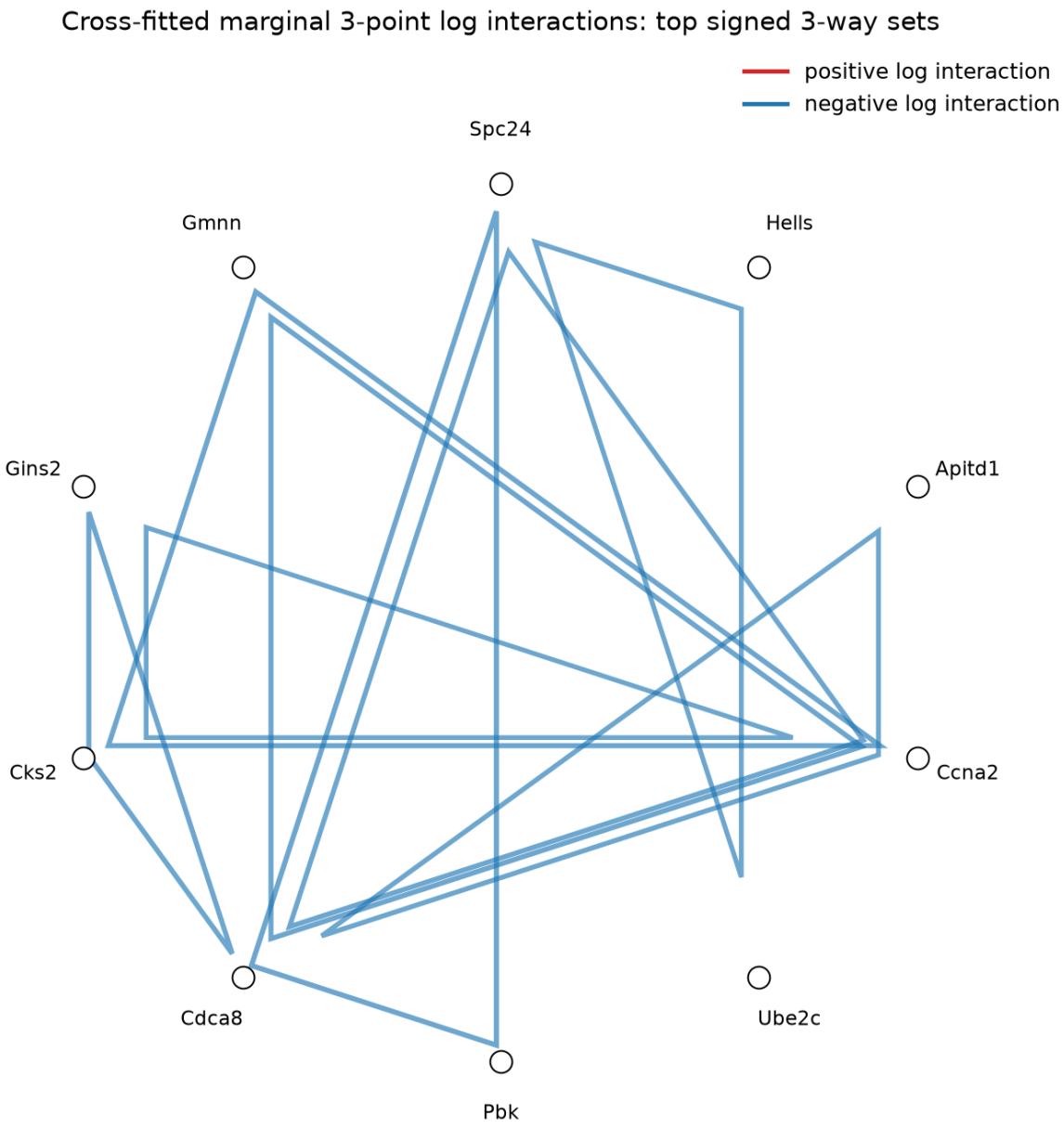


Figure 7: 12-gene marginal interactions, order 3: hypergraph.

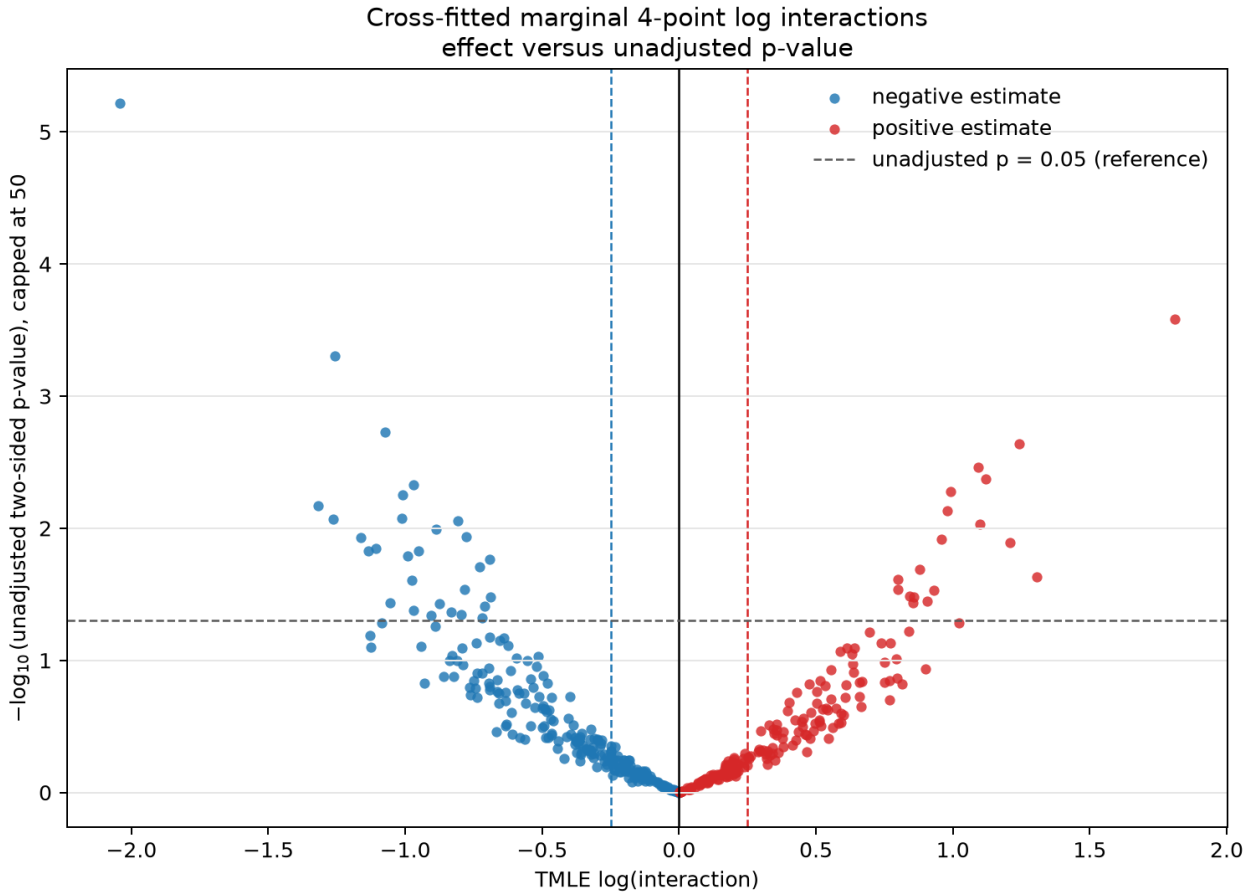


Figure 8: 12-gene marginal interactions, order 4: overview.

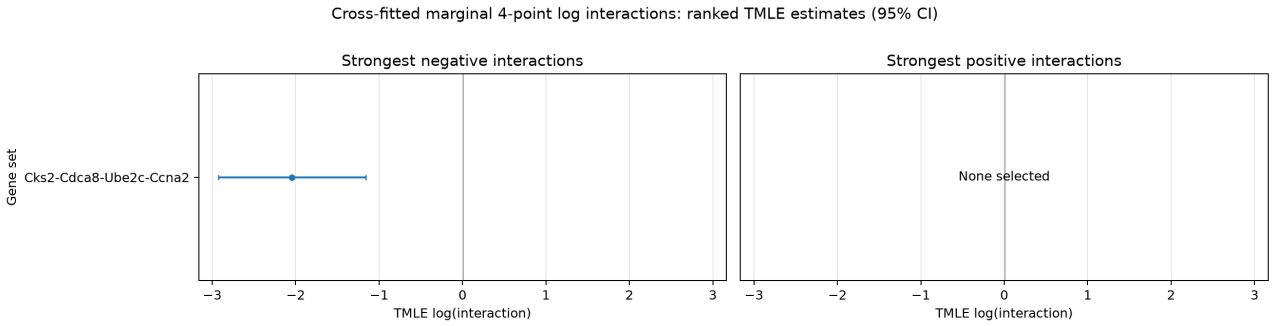


Figure 9: 12-gene marginal interactions, order 4: forest summary.

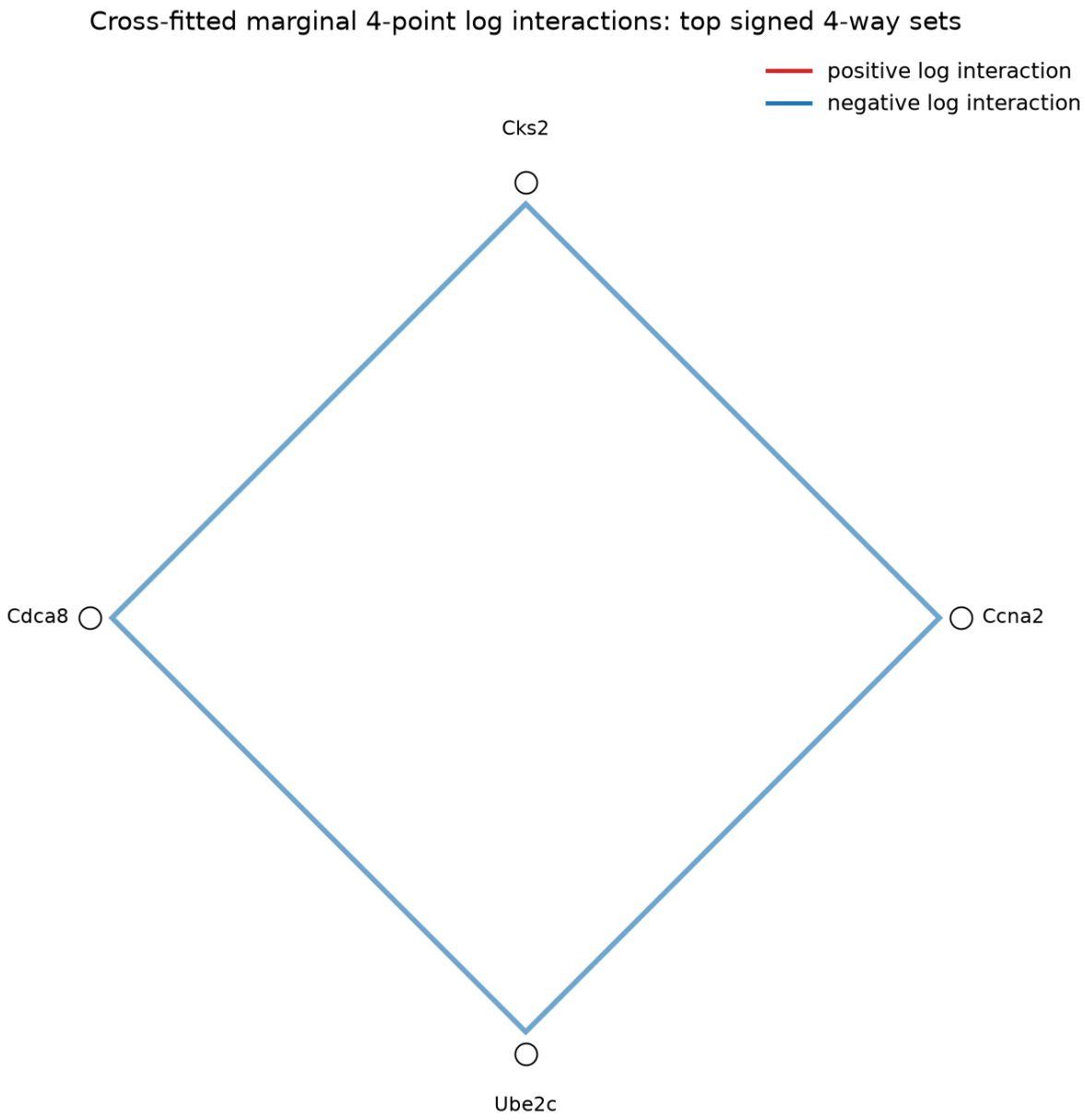


Figure 10: 12-gene marginal interactions, order 4: hypergraph.

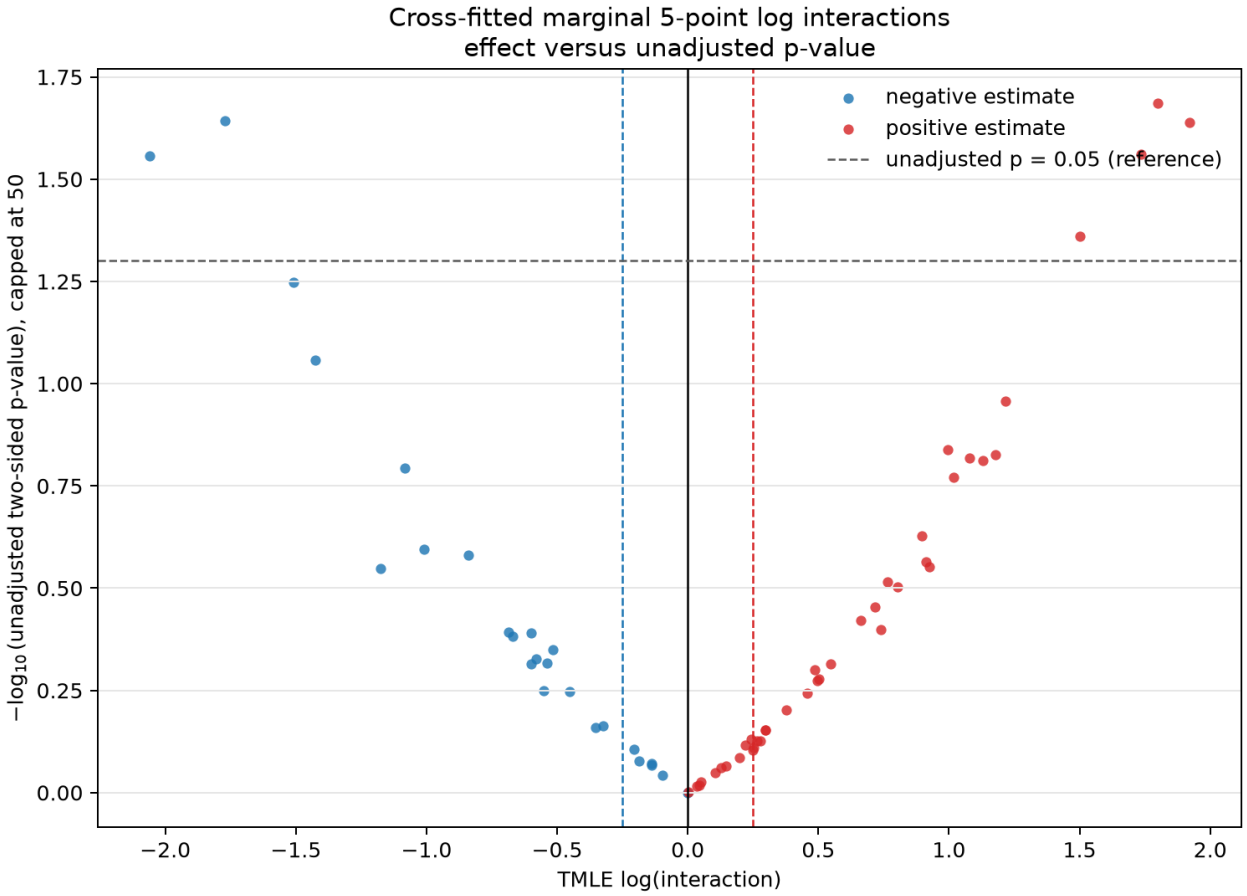


Figure 11: 12-gene marginal interactions, order 5: overview.

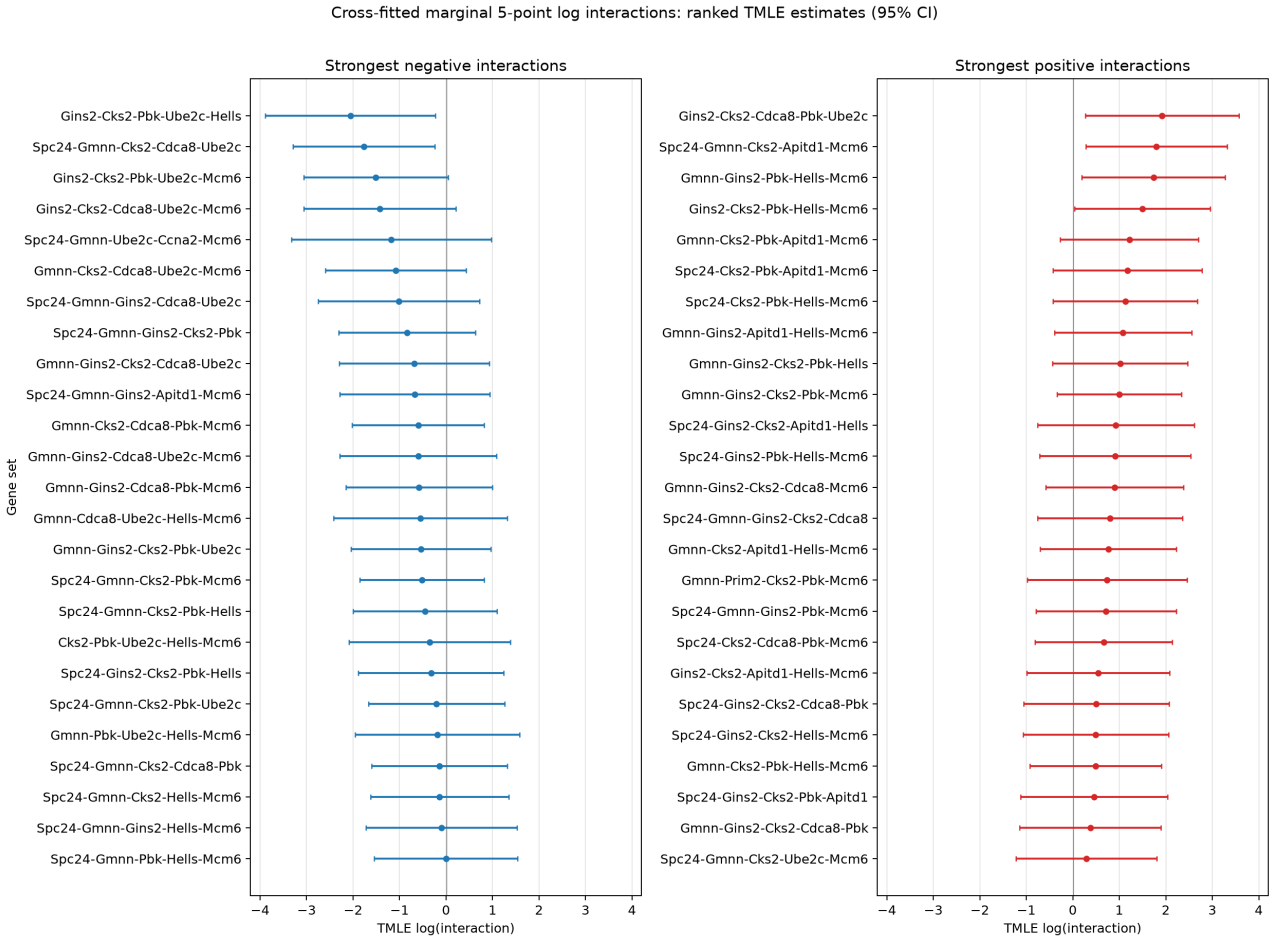


Figure 12: 12-gene marginal interactions, order 5: forest summary.

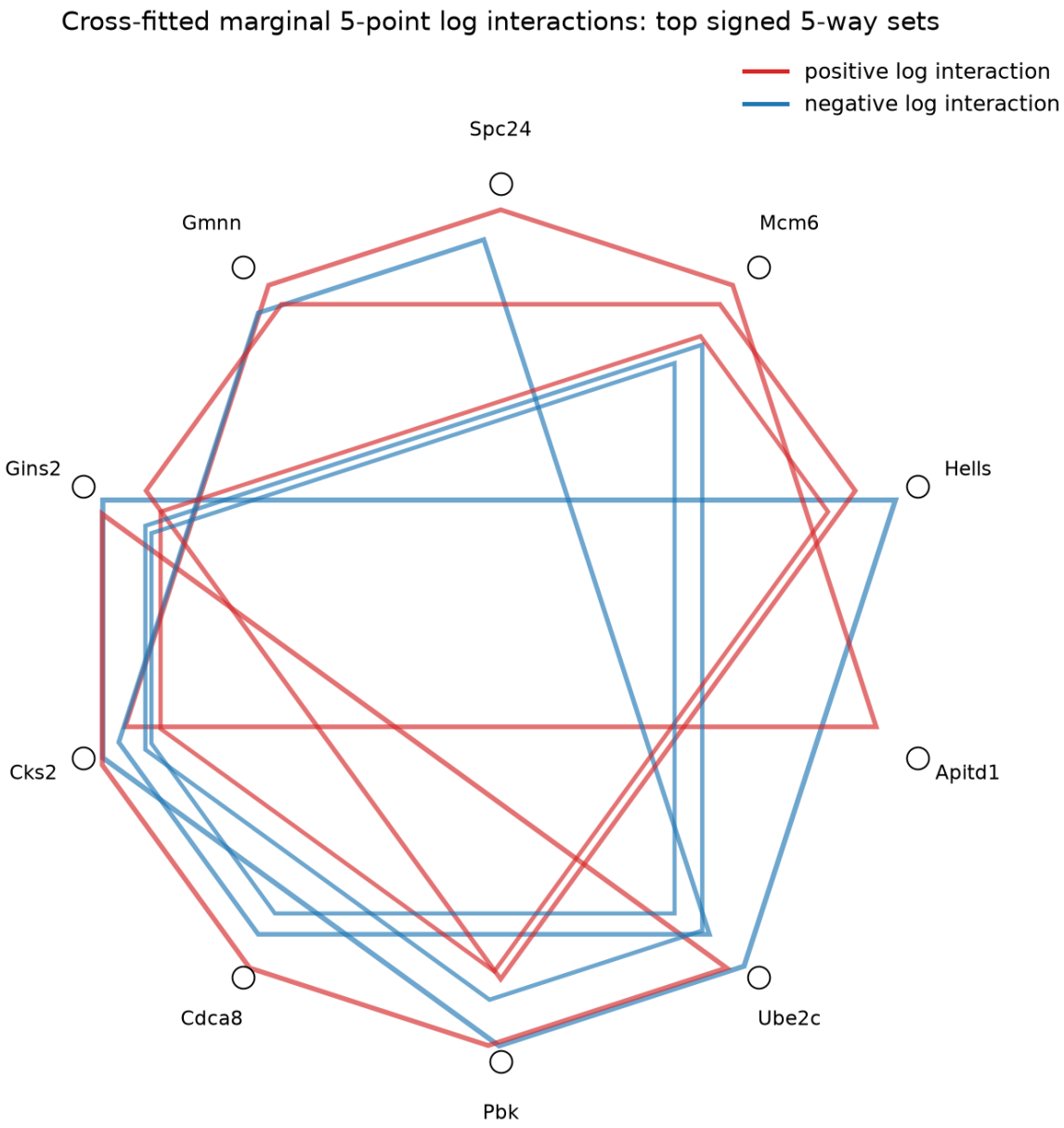


Figure 13: 12-gene marginal interactions, order 5: hypergraph.

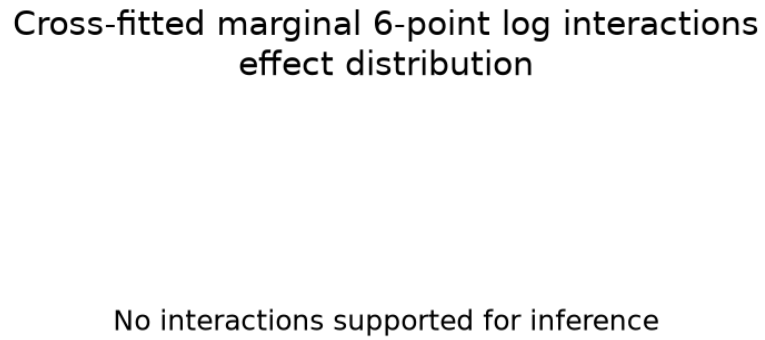


Figure 14: 12-gene marginal interactions, order 6: overview.

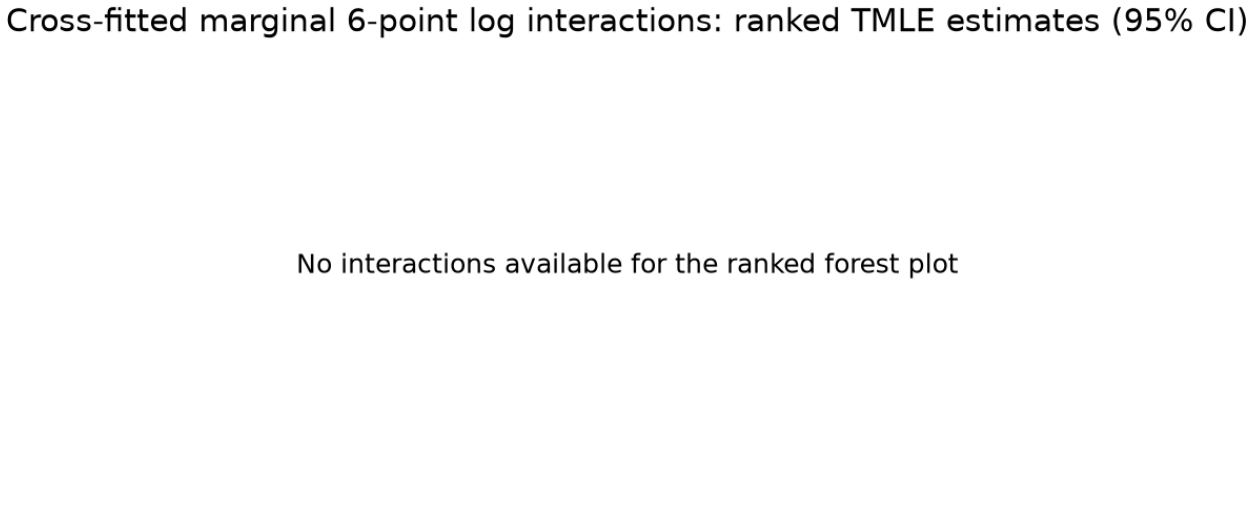


Figure 15: 12-gene marginal interactions, order 6: forest summary.

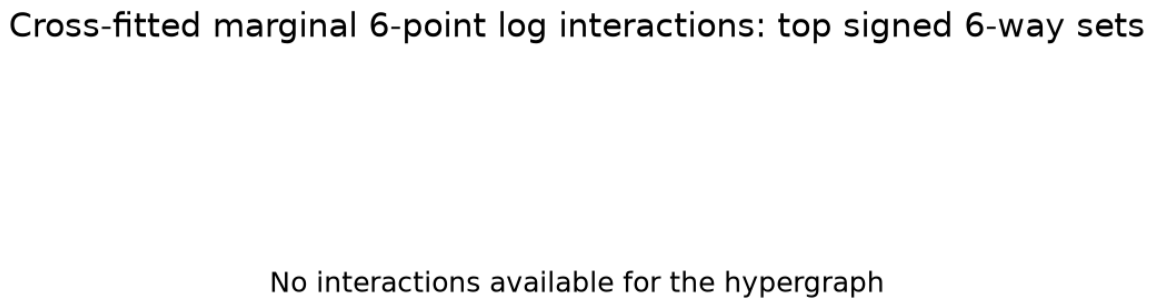


Figure 16: 12-gene marginal interactions, order 6: hypergraph.

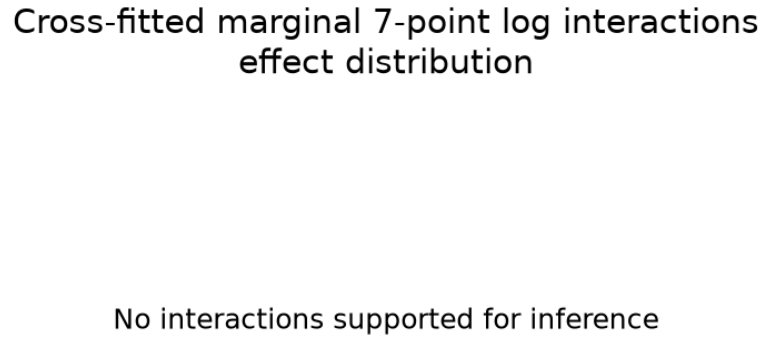


Figure 17: 12-gene marginal interactions, order 7: overview.

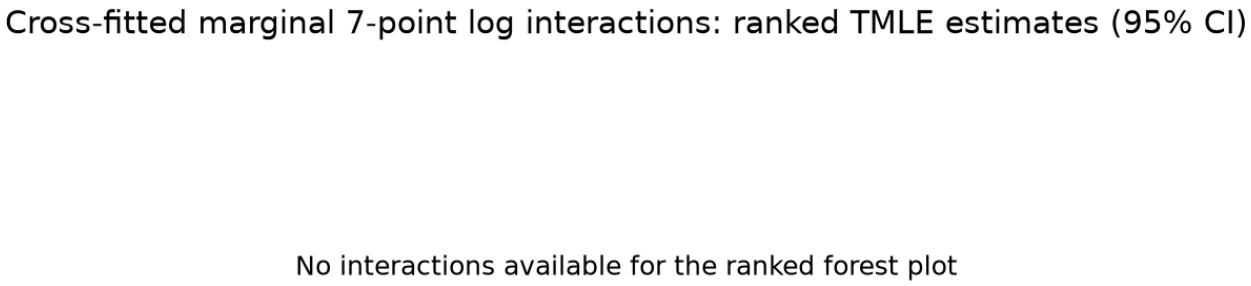


Figure 18: 12-gene marginal interactions, order 7: forest summary.

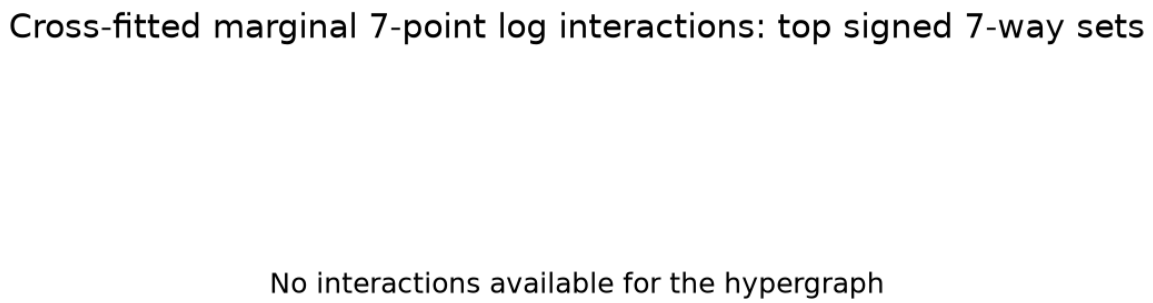


Figure 19: 12-gene marginal interactions, order 7: hypergraph.

8 Results: 999-gene study

The all-pairs run constructs 498,501 four-cell tables from its exact cache. The fresh sequential pipeline took 15.66 s: 12.16 s fresh training plus 3.50 s for a forced cache rebuild, all-pairs TMLE, results, and figures. With `minimum-cell-count=0` and `minimum-probability=1e-12`, 389,779 pairs were finite, targeted, and supported; 108,722 did not have a finite targeted estimate in both folds. BH was therefore calculated over 389,779 supported pairs; 132,177 pass both the default $q \leq 0.05$ and $|\theta| \geq 0.25$ summary-display cuts. The overview below displays $-\log_{10}(p)$, not $-\log_{10}(q)$, so its nominal horizontal reference is not the BH cut. Higher-order 999-gene results are explicitly exploratory because pairwise results select their candidates; exhaustive scans beyond pairs are combinatorially infeasible.

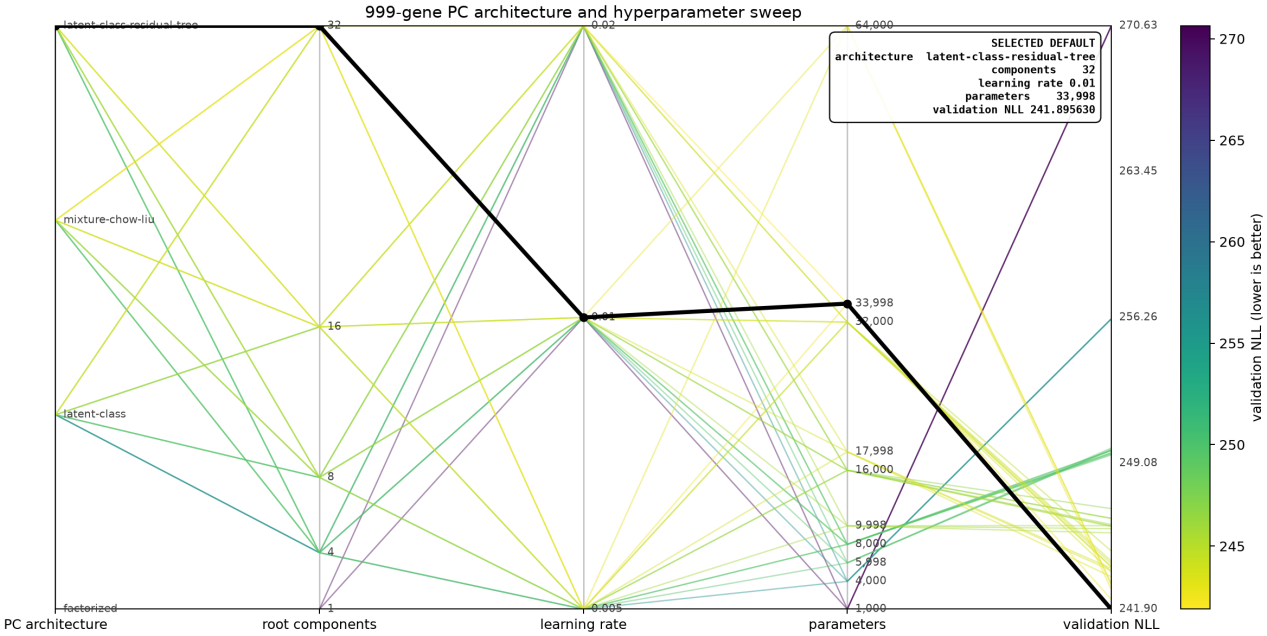


Figure 20: 999-gene PC architecture sweep retained for provenance.

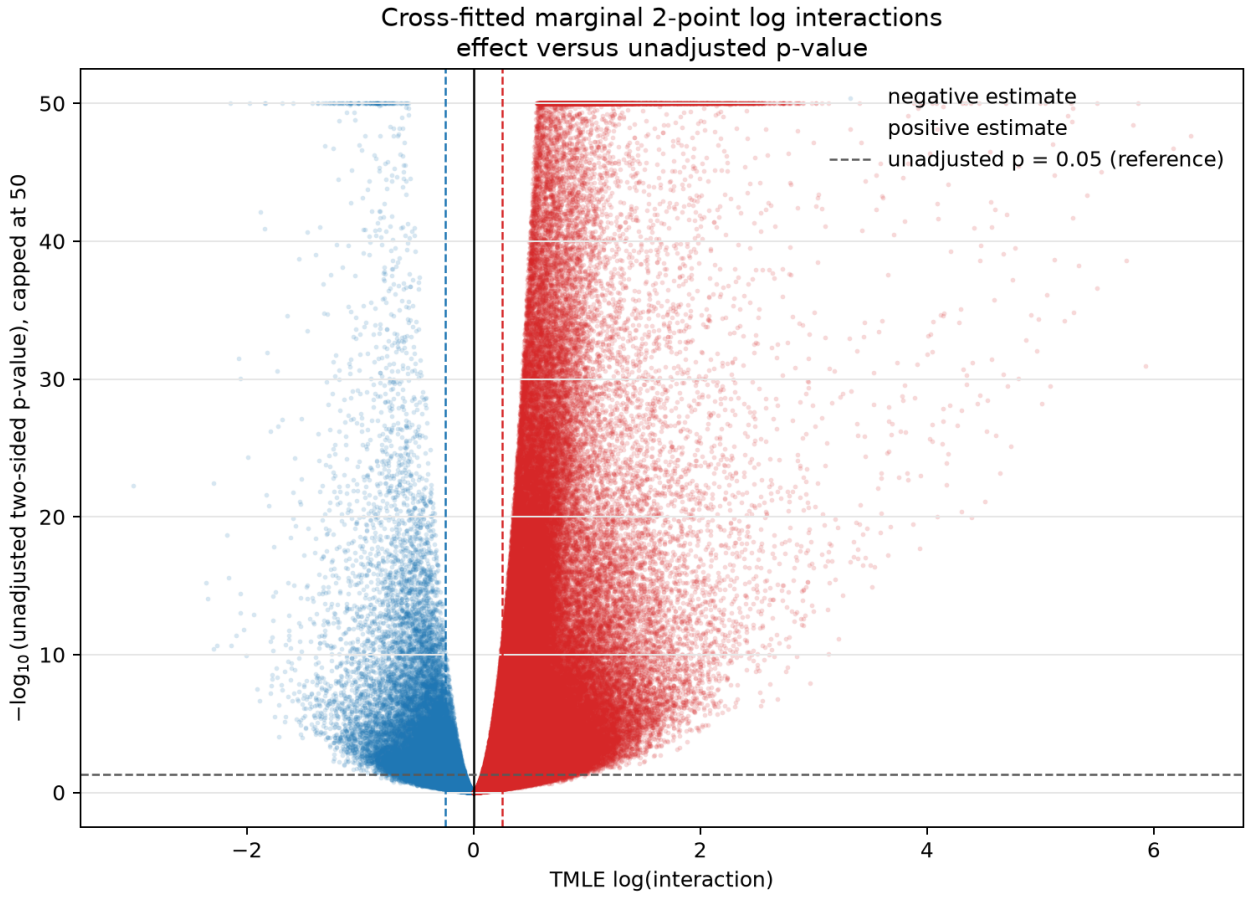


Figure 21: 999-gene marginal pairwise interactions: overview.

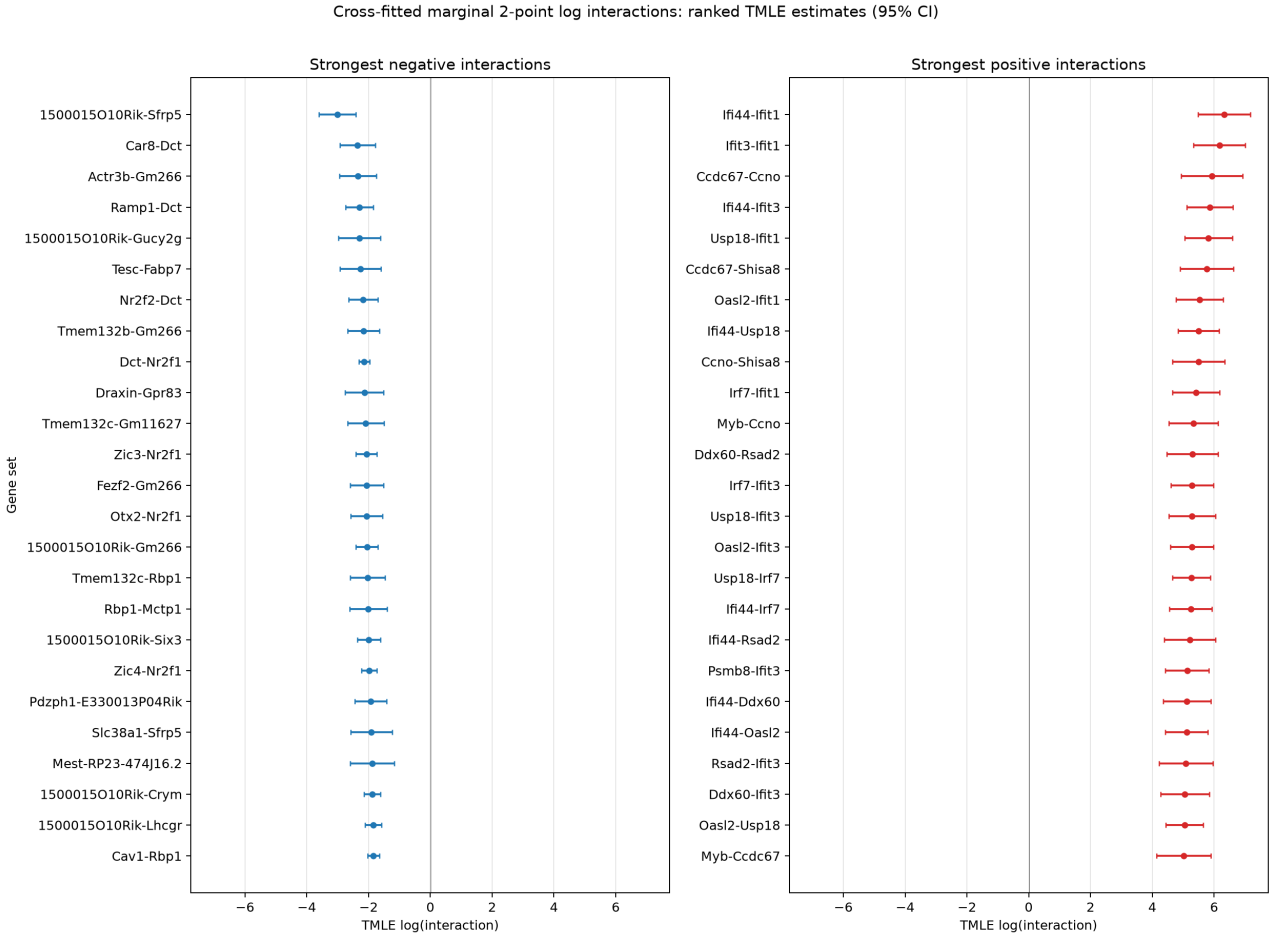


Figure 22: 999-gene marginal pairwise interactions: forest summary.

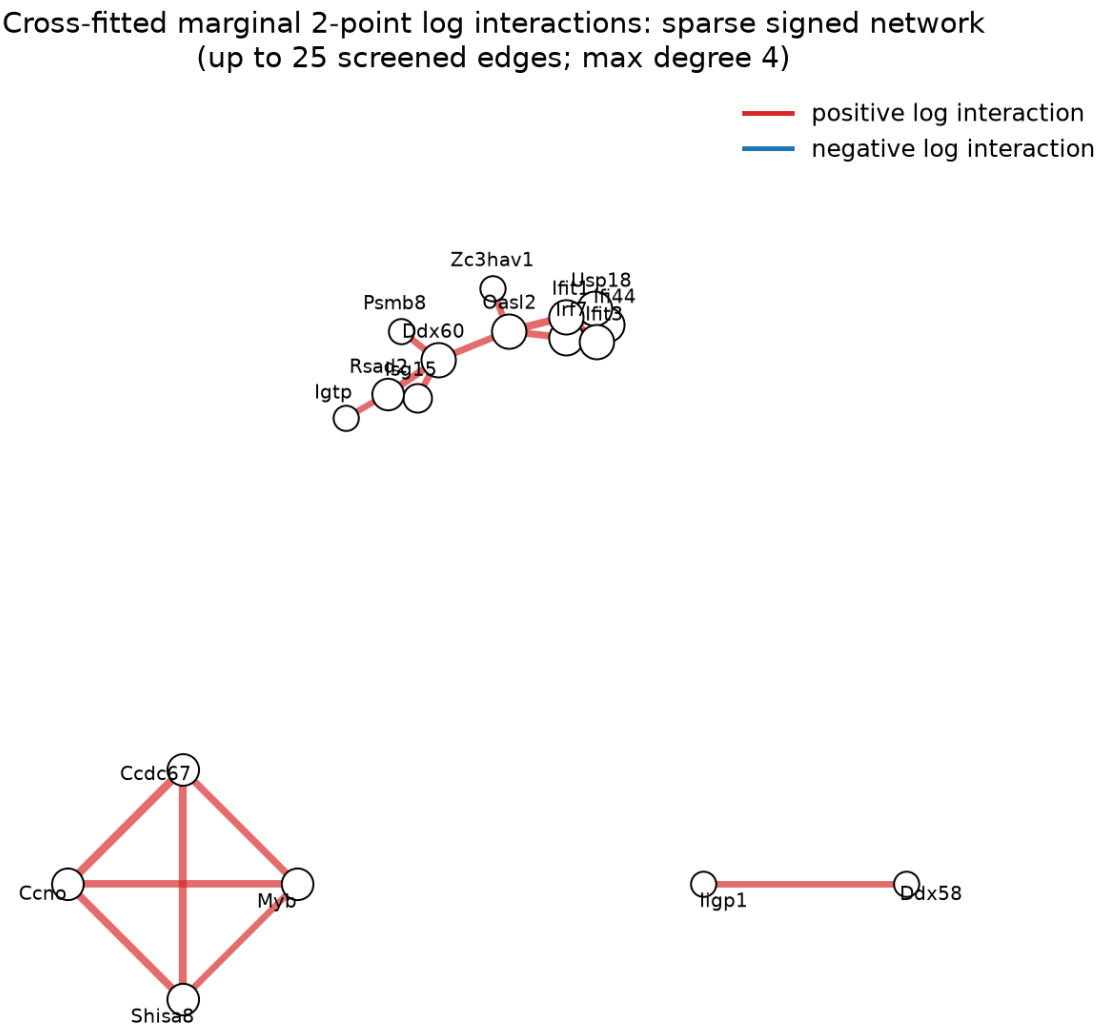


Figure 23: 999-gene marginal pairwise interactions: sparse signed network.

9 References and further reading

[1] van der Laan, M. J. and Rose, S. *Targeted Learning: Causal Inference for Observational and Experimental Data*. Springer, 2011. Background on TMLE, influence functions, and cross-validation.

[2] Pearl, J. *Causality*, 2nd ed. Cambridge University Press, 2009. A concise account of why association alone is not a causal effect.

[3] Benjamini, Y. and Hochberg, Y. "Controlling the false discovery rate: a practical and powerful approach to multiple testing." *Journal of the Royal Statistical Society: Series B* 57(1), 289–300, 1995. doi:10.1111/j.2517-6161.1995.tb02031.x.

[4] Benjamini, Y. and Yekutieli, D. "The control of the false discovery rate in multiple testing under dependency." *Annals of Statistics* 29(4), 1165–1188, 2001. doi:10.1214/aos/1013699998.